# Principal component analysis

目的：suppress redundant information，保留主要的 scene variance

used to analyze multivariate data

应用：multichannel image processing（卫星图像处理）

卫星上的传感器

不同频谱下，综合成一张图片.

3 visible light spectrum　　4 infrared and thermal bands

rectangular array of numbers

each number indicating the signal intensity

需要的背景知识：Diagonalization of Symmetric Matrices

Quadratic forms

Constrained Optimization

SVD.

The "multidimensional" character of the data refers to the three spectral dimensions rather than the two spatial dimensions that naturally belong to any photograph.

空间的

PCA的第一步：中心化 使所有数据 have a zero mean.

然后计算 (sample) covariance matrix：$S = \frac{1}{N-1} BB^T$

$(BB^T)^T = BB^T$ = symmetric